

AMPEL: Adaptive Multi-Phased Embedded Traffic Light System

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Motivation - smart city - easy to setup, no cable necessary - setup for ampeln for baustelle

Prototype (2 ESP) - Dedicated Master, Slave - Hardcoded Phases Main Part: - Same code on all devices, Master orchestrates Setup, dedicated cmds - Roles - States (Phases) - Communication Diagram (Code erklären) - Config (Defines, *ampel_config_t*, *button_config_t*)

- Delay calculation (RTT) - Resilience (Failed Master/Slave; keepalives) - Switching the lights - Button pressed, interrupts

System properties - Hardware - Range

Outlook: - Fussgaenger ampeln - Expanding groups - multiple setup - additional commands - configure via build (groups, master, group size) - security (no encryption atm) - risk of jamming of signal - Master negotiation - Threads are non deterministic - config over the air

1 Introduction

Hello